

AhamX Bodhi

Where I Becomes Infinite

Level 1: Foundation Models

Module 2: Generative AI – Core Concepts & Architecture
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Session Overview

Session Details

- **Module:** 2 – Core Concepts & Architecture
- **Duration:** 10 minutes
- **Bloom's Level:** B2 (Understand)
- **Difficulty:** L2 (Intermediate)
- **Target:** Competent Practitioner

Module 2 Entry Point

This concept opens Module 2. You are transitioning from **understanding GenAI at a high level** to understanding **how foundation models work inside** – the architecture, scale, and training principles that power every modern LLM.

Learning Objectives



By the End of This Session You Will Be Able To

1. **Define** a foundation model and explain its distinguishing characteristics
2. **Describe** the self-supervised pre-training process that creates foundation models
3. **Identify** the key properties: scale, generality, emergence, and adaptability
4. **Compare** foundation models to task-specific models and classical ML
5. **Locate** major foundation models (GPT-4, Claude, Gemini, Llama) on the landscape



Prerequisites and Context

You Should Already Know

- What Generative AI is and how it differs from discriminative AI
- The basic idea of a language model (token prediction)
- The five-level GenAI capability framework
- Module 1 content and the progressive skill stack

This Session Unlocks

- Transformer architecture deep-dive (Module 2)
- Self-attention and positional encoding
- Tokenization and vocabulary construction
- Fine-tuning and adaptation methods
- Pre-training from scratch (Module 9)





Foundation Models: Core Definition

Definition

A **foundation model** is a large-scale neural network trained on broad, diverse data using self-supervised learning, producing a general-purpose representation that can be **adapted** to a wide range of downstream tasks with minimal additional training.

The Key Idea in One Sentence

Instead of training a separate model for each task, you **pre-train once on everything** and then **adapt cheaply to anything** – this is the paradigm shift that defines modern GenAI.



Four Defining Properties of Foundation Models

$\approx 20 T$ / Parameters

Scale

- Billions to trillions of parameters.
- Trained on trillions of tokens from internet-scale corpora
- Requires thousands of GPUs over weeks to months
- GPT-4: estimated $>1T$ parameters; Llama 3: 8B to 405B

Emergence

- Capabilities not explicitly trained appear at scale
- Chain-of-thought reasoning, arithmetic, analogical thinking
- Emergent abilities: unpredictable from smaller models
- A defining feature distinguishing LLMs from prior NLP

Size ?
→ Kimi 2

$1T$ Parameters
MOE .



Four Defining Properties of Foundation Models

Generality

- Not optimized for any single task during pre-training
- Learns language, reasoning, world knowledge together
- Single model handles translation, code, math, dialogue

Adaptability ,

Level 2

- Fine-tune with hundreds (not millions) of examples
- Prompt engineering: zero configuration adaptation
- LoRA / QLoRA: adapt with $<1\%$ of parameters updated
- RLHF: align behavior with human preferences

I like Cat very much



How Foundation Models Are Created: Pre-Training

Self-Supervised Learning: The Core Mechanism

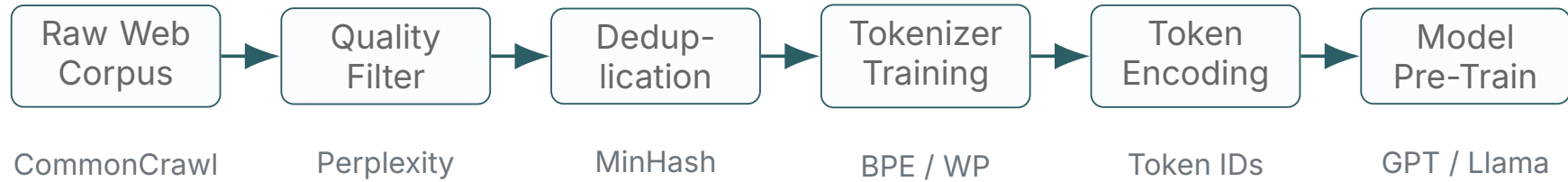
- **No human labels required:** The training signal comes from the data itself
- **Causal Language Modeling:** Predict the next token given all previous tokens
 $P(w_t | w_1, w_2, \dots, w_{t-1})$
- **Masked Language Modeling:** Predict masked tokens from surrounding context (BERT-style)
- **Loss function:** Cross-entropy over the vocabulary at each position
- **Gradient descent** over billions of steps updates all parameters to minimize loss

Why This Works So Well

Predicting the next word **forces the model to learn grammar, facts, reasoning, and style** simultaneously. There is no shortcut – the only way to predict well is to truly understand.



Pre-Training Data Pipeline



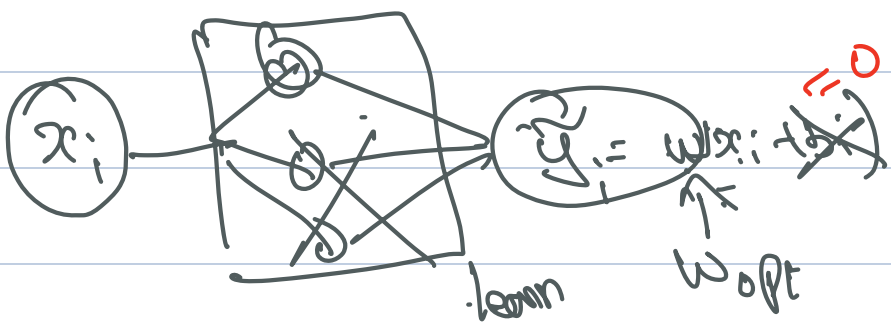
Training data quality is as important as model architecture. Modern foundation models filter for quality, deduplicate aggressively, and balance domain mix before a single gradient step is taken.

$$e_{\min} = || \hat{f}(w) - f^{\text{actin}} ||_0$$

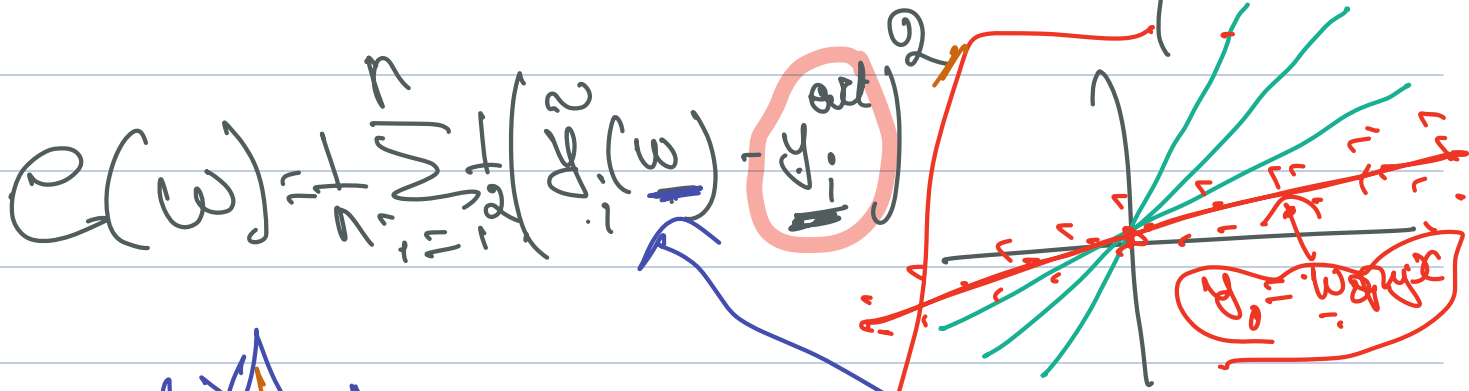
$$\rightarrow y(x_i) = w x_i + b \rightarrow y_i^{\text{act}}$$

Daten

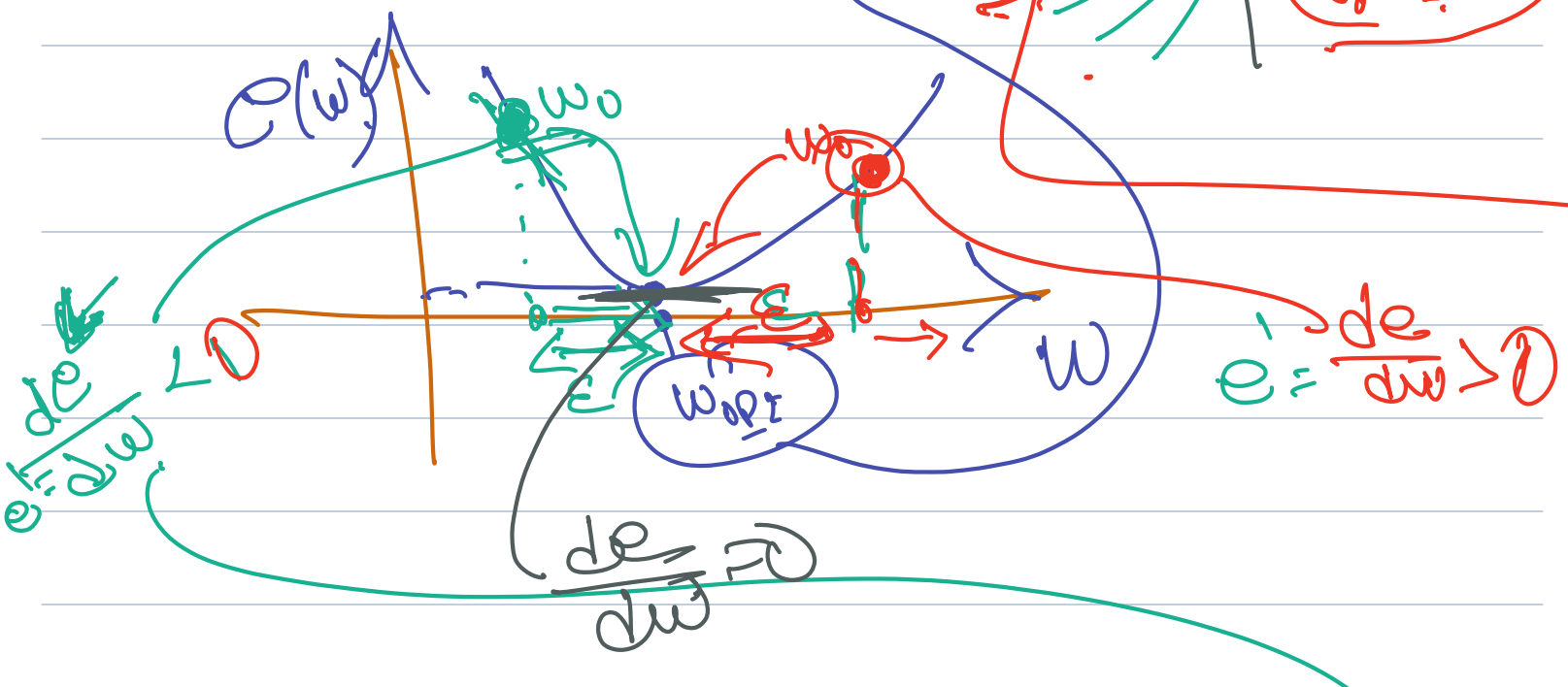
x_i	y_i^{exat}
x_1	y_1
$\vdots$	$\vdots$
x_n	y_n



$$x_{\text{conc}} \rightarrow y$$



$$e(w) = \frac{1}{n} \sum_{i=1}^n \frac{1}{2} (\hat{y}_i(w) - y_i^{\text{act}})^2$$



$$w^{opt} = w^0 + \epsilon = w^0 - \epsilon \text{sign}(e')$$

$$w^{opt} = w^0 - \epsilon = w^0 - \underline{\underline{\epsilon \text{sign}(e')}}$$

$$\begin{pmatrix} w_1^{opt} \\ \vdots \\ w_n^{opt} \end{pmatrix} = \begin{pmatrix} w_1^0 \\ \vdots \\ w_n^0 \end{pmatrix} - \eta \begin{pmatrix} \frac{de}{dw_1} \\ \vdots \\ \frac{de}{dw_n} \end{pmatrix}$$

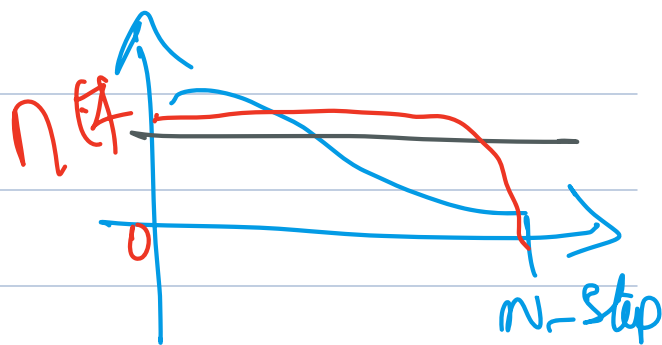
$$W^{k+1} = W^k - \eta \nabla E$$

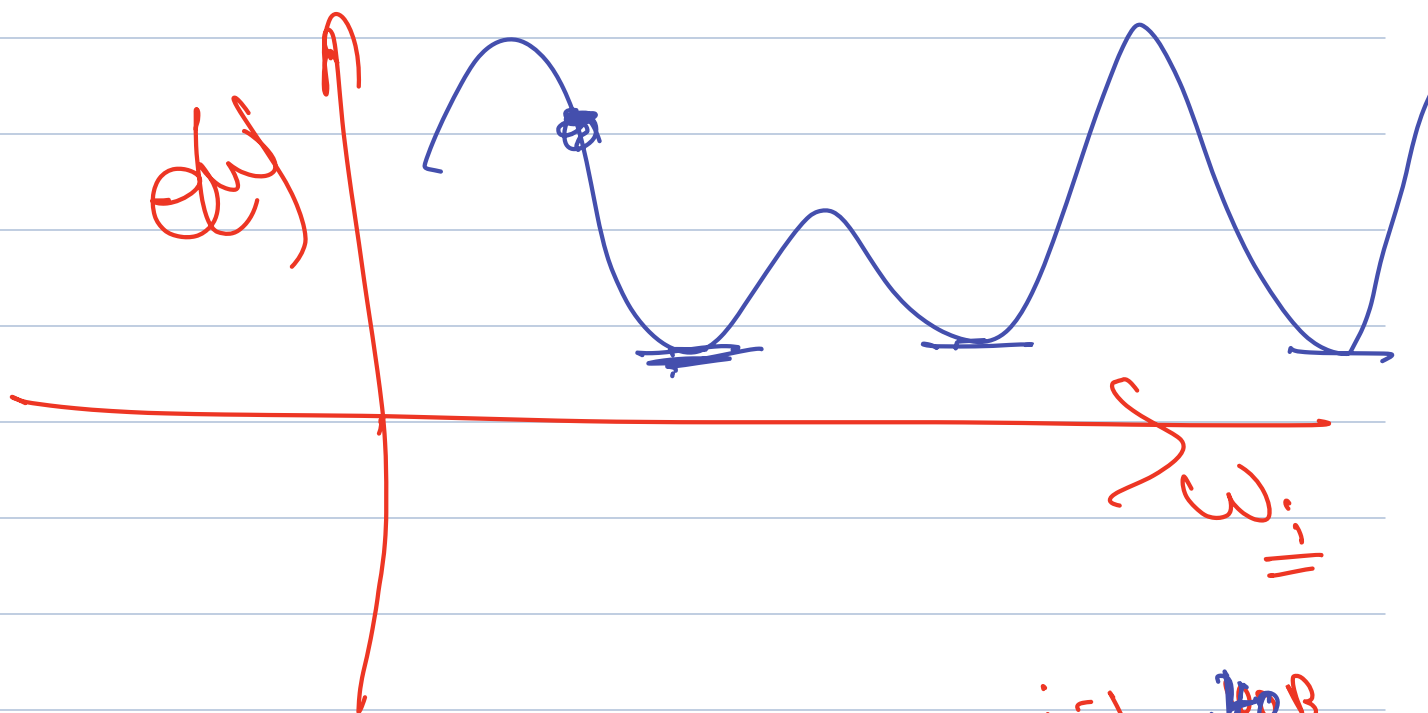
GD -

learning rate

I like cat very much

$P_1 \rightarrow$ I ~~like~~ cat
 $P_2 \rightarrow$ I like ~~cat~~





$i = 1, \dots, N$



Foundation Models vs. Prior Paradigms

Attribute	Classical ML	Task-Specific DL	Foundation Model
Training data	Hundreds to thousands	Thousands to millions	Billions to trillions
Parameters	Thousands	Millions	Billions to trillions
Task coverage	One task, fixed	One task, fixed	Many tasks, general
Adaptation cost	Retrain from scratch	Fine-tune entire model	Prompt or few-shot
Emergent abilities	None	Rare, domain-specific	Yes – new at scale
Examples	SVM, XGBoost	ResNet, BERT-base	GPT-4, Claude, Llama 3



The Foundation Model Landscape (2024--2025)

Closed / API-Access Models

- **OpenAI:** GPT-4o, GPT-4-Turbo, o1, o3
- **Anthropic:** Claude 3.5 Sonnet, Claude 3 Opus
- **Google:** Gemini 1.5 Pro, Gemini Ultra
- Access via REST API; weights not released
- Best quality; highest cost per token

Open-Weight Models

- **Meta:** Llama 3.1 (8B, 70B, 405B)
- **Mistral AI:** Mistral 7B, Mixtral 8×22B
- **Microsoft:** Phi-3, Phi-4 (small but capable)
- **TII:** Falcon 180B
- Weights downloadable; self-hostable; fine-tunable





Multimodal Foundation Models

What Is Multimodality?

A **multimodal foundation model** accepts and/or generates multiple types of data: text, images, audio, video, and code within a single unified model architecture. The Transformer backbone is extended with modality-specific encoders and decoders.

Key Multimodal Systems

- **GPT-4o:** Text, image, and audio input; text and audio output (real-time)
- **Gemini 1.5 Pro:** Text, image, audio, video, and code – 1M token context
- **Claude 3.5 Sonnet:** Text and image input with strong document understanding
- **LLaVA / InternVL:** Open-weight vision-language models for research
- **Whisper:** Audio-to-text foundation model from OpenAI (open-weight)





Application: Conversational AI & Assistants

What Foundation Models Enable

- Context-aware, multi-turn dialogue across any topic
- System prompt controls persona, tone, and task constraints
- Function calling enables the model to invoke external tools
- Memory augmentation extends context beyond the context window

Production Deployments

- **ChatGPT (OpenAI):** 100M+ daily active users; GPT-4o backbone
- **Claude.ai (Anthropic):** Long-context document work; 200K token window
- **Gemini (Google):** Deep Search integration; multimodal queries
- **Copilot (Microsoft):** Enterprise productivity embedded in Office 365



Application: Code Generation & AI-Assisted Development

Capabilities

- Autocomplete at function and file level
- Natural language to working code (NL2Code)
- Bug detection, explanation, and test generation
- Cross-language translation (Python → C++)
- Repo-level context for complex refactors

Industry Tools

- **GitHub Copilot:** Claude/GPT-4 backbone, 1.8M paying users
- **Cursor:** Codebase-aware AI IDE; Sonnet 3.5 backend
- **Claude Code:** Agentic terminal coding (Anthropic)
- **Gemini Code Assist:** Google Cloud integration
- **Devin:** Autonomous software engineering agent



Application: Enterprise Search & Retrieval-Augmented Generation

The Problem Foundation Models Solve

- Enterprises have petabytes of internal documents with no semantic search
- Foundation models convert documents and queries into dense vector representations
- Similarity search retrieves relevant passages; the model generates grounded answers
- Result: accurate, cited answers that reduce hallucination

Production Stacks

- **Vector databases:** Pinecone, Weaviate, Qdrant, pgvector
- **Orchestration:** LangChain, LlamaIndex, Haystack
- **Embedding models:** text-embedding-3-large (OpenAI), E5-Mistral
- **LLM backend:** GPT-4o, Claude 3.5, Llama 3 70B (self-hosted)



Application: Scientific AI & Research Acceleration

Life Sciences

- **AlphaFold 3:** Protein and molecular structure prediction
- **ESM-3 (Meta):** Protein language model – 3B parameters
- **Med-PaLM 2:** Expert-level medical question answering
- **BioGPT:** Biomedical literature generation and QA

Physical Sciences

- **FourCastNet / Pangu:** Global weather forecasting LLMs
- **Aurora (Microsoft):** Foundation model for Earth systems
- **GraphCast (DeepMind):** 10-day weather at 0.25-degree resolution
- **MatBERT:** Foundation model for materials science



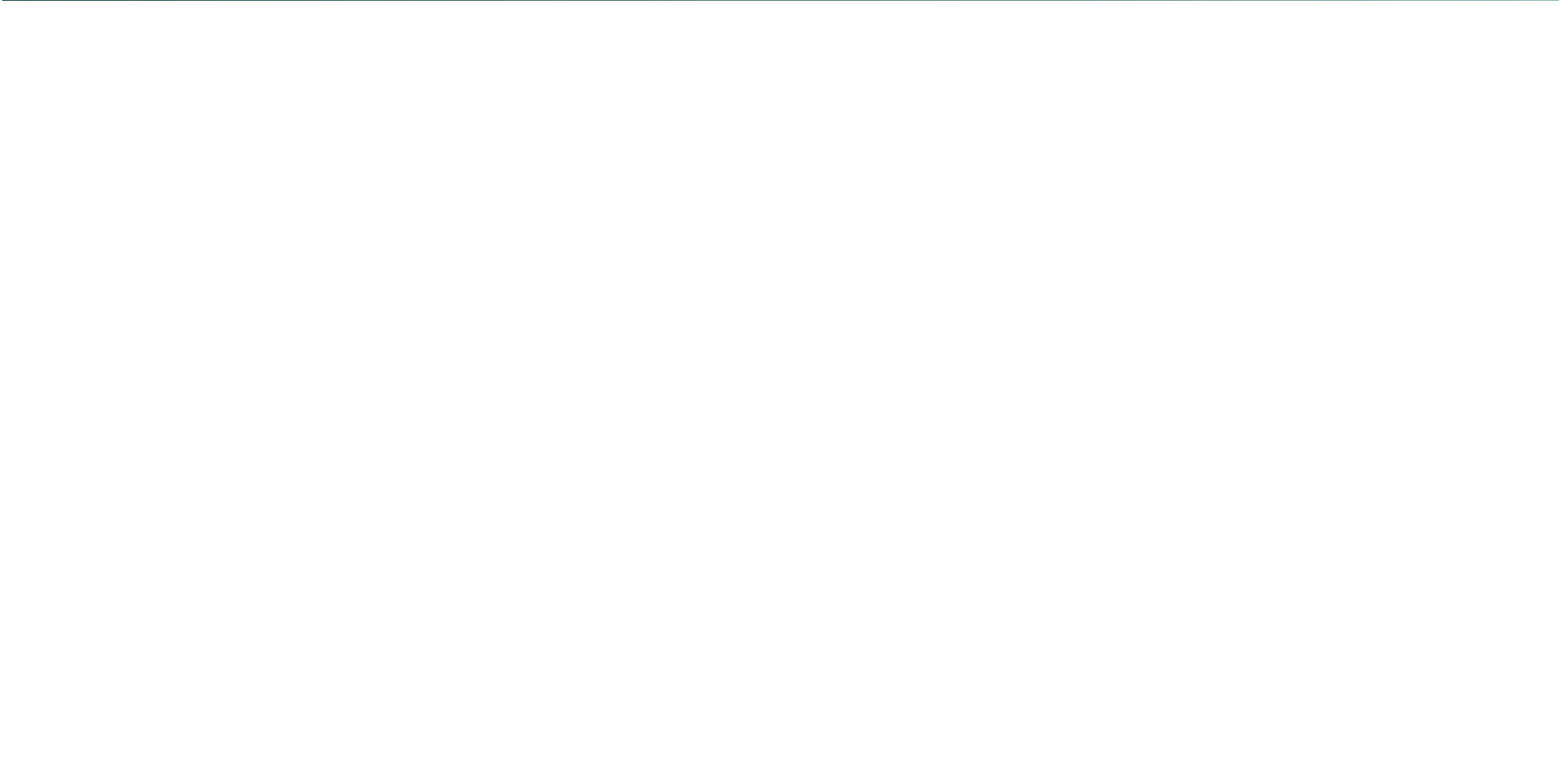
Application: Foundation Models as Agent Brains

The Agent Architecture

In agentic systems, the foundation model is the **reasoning engine**. It receives observations from the environment, decides which tools to call, executes multi-step plans, and reflects on outcomes to correct errors. The model's broad pre-training knowledge enables it to generalise to novel tasks without task-specific re-training.

Production Agentic Systems

- **AutoGPT / BabyAGI:** Early autonomous agents on GPT-4
- **OpenAI Assistants API:** Tool use, file retrieval, code interpreter
- **Anthropic Claude with MCP:** Model Context Protocol for tool integration
- **LangGraph:** Multi-agent orchestration with state machines



Skills You Are Building



Conceptual Skills

- Explaining what a foundation model is to technical and business audiences
- Comparing open-weight vs. closed API models for a given use case
- Understanding scaling laws and emergent capability thresholds
- Mapping GenAI tasks to appropriate model families

Applied Skills

- Selecting the right foundation model for a task (cost / quality / privacy)
- Identifying when fine-tuning vs. prompting is appropriate
- Reading model cards and understanding benchmark results
- Evaluating model capabilities using standardized benchmarks (MMLU, HumanEval)



Professional Role Applications

Role	How Foundation Model Knowledge Is Applied
AI / ML Engineer	Architecture selection, deployment, API integration, benchmarking
Data Scientist	Pre-training data curation, evaluation, bias analysis, fine-tuning
Software Developer	API integration, prompt design, function calling, cost optimisation
Research Scientist	Novel pre-training objectives, scaling experiments, emergent ability studies
Product Manager	Capability scoping, model selection, cost-benefit tradeoffs, roadmap
MLOps Engineer	Serving infrastructure, quantisation, batching, latency optimisation



Knowledge Graph: What You Needed to Get Here

Prerequisite Concepts (Module 1)

- **What is Generative AI?** — The paradigm-level definition and motivation
- **Generative vs. Discriminative Models** — Mathematical foundations $P(x)$ vs $P(y|x)$
- **Five-Level Framework for GenAI** — Where foundation models fit in the ecosystem
- **Progressive Skill Stack** — Why Tier 1 is mastering foundation model internals
- **From Using to Building Models** — The practitioner mindset shift this concept supports



Knowledge Graph: What This Concept Unlocks

Immediate Unlocks (Module 2)

- Transformer Architecture Overview
- Self-Attention Mechanism
- Positional Encoding
- Tokenization: BPE and WordPiece
- Context Windows and KV Cache

Downstream Unlocks

- Prompt Engineering Principles (Module 2)
- What is Fine-Tuning? (Module 7)
- LoRA and QLoRA (Module 7)
- Pre-Training from Scratch (Module 9)
- LLMOps and Serving (Module 9)



Common Misconceptions to Overcome

Critical Corrections

- **"Foundation models understand language"** – They are statistical predictors of token sequences; understanding is a loaded term requiring careful qualification
- **"Bigger is always better"** – Phi-3 Mini (3.8B) outperforms some 70B models on reasoning when trained on high-quality curated data (quality over quantity)
- **"Foundation models are lookup tables"** – Generalisation to novel inputs demonstrates systematic compositional reasoning not reducible to memorisation
- **"Open-weight means open-source"** – Llama 3 weights are available but Meta restricts commercial use above 700M monthly active users; check the license
- **"Pre-training is fine-tuning at larger scale"** – They differ in data, objective, learning rate schedule, and compute by orders of magnitude

Session Summary



Key Takeaways

- Foundation models are pre-trained at scale on diverse data using self-supervised learning
- Four properties define them: scale, generality, emergence, adaptability
- They replaced task-specific models as the dominant AI paradigm from 2020 onward
- Open-weight and closed-API models serve different cost-privacy-quality trade-offs

What Comes Next

- **Next concept:** Transformer Architecture Overview
- You will learn the encoder-decoder structure, attention heads, and how the architecture enables next-token prediction at massive scale
- **Lab:** Loading and querying a foundation model via the Hugging Face API
- Start preparing your development environment if you have not already

Thank You

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